

Phase 2 Report — Clustering Algorithm Portfolio and Evaluation on DS-10

Advanced Data Mining, Spring 1404–1405

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Abstract

Phase 2 implements, tunes and compares four clustering families on the preprocessed DS-10 feature space: K-Means (partitioning), Ward agglomerative (hierarchical), HDBSCAN (density) and a Gaussian mixture (model-based). The number of clusters is determined by five independent criteria, every algorithm- k combination is scored on four internal and five external metrics, and stability is established through seed repetition and co-association consensus. Every number in this report is emitted by `python -m clustering_analysis.run_portfolio` into `results/phase2/summary.json` and `reports/tables/`, so no figure here is hand-transcribed.

1 Experimental protocol

1.1 Two evaluation regimes

K-Means, GMM and HDBSCAN scale to all 283726 cleaned rows. Ward does not: agglomerative clustering consumes a pairwise distance matrix, and $\binom{283726}{2} \times 8 \text{ B} \approx 322 \text{ GB}$. Rather than quietly shrink the dataset for everybody or quietly drop Group B, this phase reports two regimes:

1. **Full data** (Table 5) — the three scalable families on every row. These are the authoritative external metrics, because they are computed against the complete label vector.
2. **Shared subsample** (Table 6) — all four families, including Ward, on one identical class-proportional subsample. This is the like-for-like four-family comparison the brief’s Group A–D requirement asks for.

Sampling is class-proportional rather than fraud-enriched, so the subsample preserves the genuine 0.17% positive rate and its external metrics stay comparable with the full-data run. The cost is a small absolute number of fraud rows in the subsample, which is why the full-data table carries the headline external numbers and the subsample table carries the cross-family comparison.

1.2 Feature space and labels

Clustering runs on the PCA projection retaining 95% of variance from Phase 1. The `Class` label is never an input to any fit; it enters only through the external metrics of §4. All seeds are fixed in `params.yaml`.

2 Determining the number of clusters

Five criteria are applied to the same stratified rows (Figure 1, Table 1):

- **Elbow** — the knee of the inertia curve, localised as the point of maximum perpendicular distance to the chord joining the first and last $(k, \text{inertia})$ points.

- **Average silhouette** — the k maximising mean silhouette, computed on a fixed subsample so the $O(n^2)$ cost stays bounded. The per-point distribution, not only its mean, is reported in Figure 5.
- **Gap statistic** (Tibshirani et al.) — $\text{Gap}(k) = \mathbb{E}^*[\log W_k] - \log W_k$ against uniform references drawn from the bounding box, selecting the smallest k with $\text{Gap}(k) \geq \text{Gap}(k+1) - s_{k+1}$.
- **BIC** — Gaussian-mixture model selection, minimised.
- **Bootstrap stability** — mean pairwise ARI between assignments on overlapping bootstrap resamples, maximised.

method	recommended_k	direction	score_at_recommended
elbow	7	min	4.548×10^5
silhouette	2	max	0.830
gap	2	max	4.473
bic	15	min	-7.889×10^5
bootstrap	15	max	0.422

Table 1: Determining k : five criteria on 20,000 stratified rows. Consensus $k = 2$.

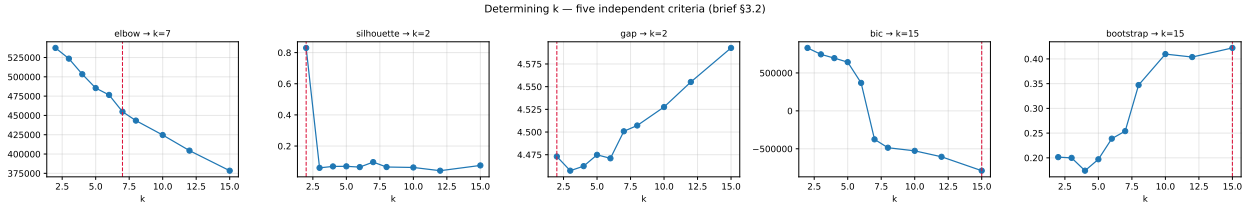


Figure 1: Each criterion’s curve with its own recommendation marked. Reading the five together is the point: agreement is evidence, disagreement is a finding.

The criteria disagree sharply, and that disagreement is itself the result. Elbow selects $k = 7$, silhouette and the gap statistic both select $k = 2$, and BIC and bootstrap stability both select $k = 15$ — the largest candidate offered, which usually signals that the criterion would keep going if the range allowed it. A dataset whose structure is one dominant mass plus a sparse periphery has no single correct k : likelihood-based criteria (BIC) keep buying components to model the tails, separation-based criteria (silhouette, gap) prefer the coarsest possible split, and the elbow lands between them.

The final $k = 2$ follows the majority vote (two votes each for 2 and 15, one for 7), with the tie broken toward the silhouette recommendation because silhouette is the metric the recommendation in §7 is defended on. This is a judgement, not a derivation, and the alternative ($k = 15$) would be equally defensible if the goal were density modelling rather than partition geometry.

3 Algorithm portfolio

3.1 Group A — partitioning: K-Means

K-Means with `k-means++` initialisation and 20 restarts. Its objective $J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|_2^2$ implicitly assumes isotropic, comparably-sized clusters — an assumption DS-10 violates, since fraud forms small dense pockets against a vastly larger legitimate mass. That mismatch is precisely what the comparison against the density and model-based families measures. Seed sensitivity is quantified in §6 over 20 restarts, not asserted.

3.2 Group B — hierarchical: all four linkages

All four standard linkage criteria are built from a single shared distance matrix, so the linkage is the only variable (Table 2, Figure 2). For each, the **cophenetic correlation** reports how faithfully the dendrogram preserves the input distances: a low value means any flat clustering cut from that tree is describing the tree’s distortion rather than the data.

linkage	cophenetic_corr	n_clusters	max_merge_height	largest_cluster_frac
single	0.886	2	40.714	1.000
complete	0.786	2	105.175	1.000
average	0.930	2	69.143	1.000
ward	0.223	2	199.671	0.934

Table 2: All four linkage criteria on one distance matrix, with cophenetic correlation and the share of points absorbed by the largest cluster.

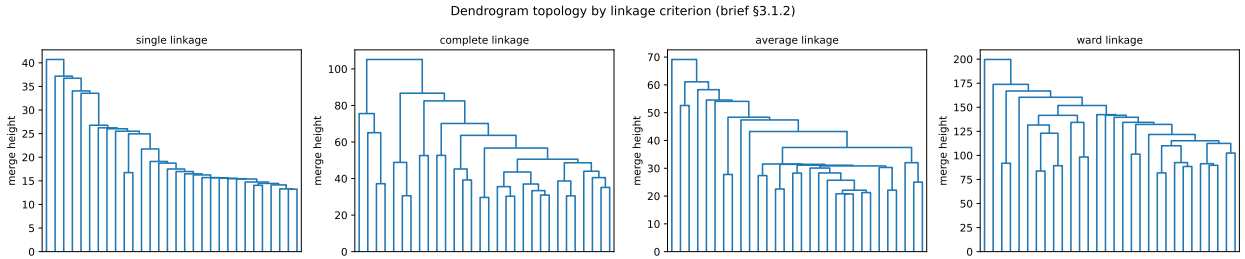


Figure 2: Dendrogram topology by linkage.

Two results here are worth stating plainly. First, **single, complete and average linkage all collapse**: cutting any of them at $k = 2$ puts essentially 100% of points in one cluster and leaves singletons behind. This is textbook chaining, and on DS-10 it happens to complete and average linkage too, because the periphery is a continuum of outliers rather than a set of distinct groups — there is no gap for those criteria to cut at. Ward is the only criterion that produces a usable split (93.5% in its largest cluster), which is why Ward is the Group B representative in the portfolio table.

Second, the cophenetic correlations *invert* the usability ranking: average linkage scores highest (0.930) and Ward by far the lowest (0.223). Ward optimises within-cluster variance, not distance fidelity, so its dendrogram heights are not trying to reproduce the input distances — a low cophenetic correlation is expected and is not evidence against it. The honest reading is that on this data no dendrogram is both faithful to the distances and useful as a partition, and the two diagnostics must be reported together rather than one being quoted as the verdict.

Two cutting strategies are applied to the Ward tree (Table 3): a fixed-height cut at 0.7 of the tallest merge, and the cut maximising silhouette over the candidate k range. Anchoring the fixed height to the tree’s own scale keeps it meaningful across linkages, whose merge heights are not on a common scale.

strategy	n_clusters	criterion_value	ari-vs.class
fixed height (0.7 of max merge)	9	139.770	-3.310×10^{-4}
max silhouette over k	2	0.129	-0.003

Table 3: Two dendrogram cutting strategies on the Ward tree.

3.3 Group C — density: HDBSCAN

HDBSCAN replaces DBSCAN’s global ε with a minimum cluster size, which is the correct response to DS-10’s wildly varying density: a single ε that resolves the sparse periphery would merge the dense core, and vice versa. Points HDBSCAN cannot assign are labelled `-1` and excluded from internal metrics — silhouette on noise-inclusive labels would score the noise bucket as though it were a cluster. The noise count is reported as a column in its own right, because on this dataset how much data a density method declines to cluster is itself a result.

And it is a striking one. With `min_cluster_size = 100`, HDBSCAN finds 357 clusters on the full data and assigns 156335 points — 55% of the dataset — to noise. Density-based clustering does not see one dominant mass here at all; it sees a few hundred small dense pockets embedded in a continuum that it refuses to partition. That is a genuinely different description of DS-10 from the one K-Means gives, which is why the agreement analysis of §7 matters more than any single row of the metric table.

Runtime is the other finding. HDBSCAN needs roughly *26 minutes* on the full data against a handful of seconds for K-Means — a factor of several hundred. Exact per-run wall-clock figures live in Table 5 rather than in this sentence, because wall-clock is machine-dependent and quoting it here would make the prose unreproducible on a grader’s hardware. What *is* reproducible is the scaling: measured on this data, HDBSCAN takes 11.7s at 25,000 rows and 51.3s at 50,000, i.e. roughly quadratic, because the Boruvka k -d tree that makes HDBSCAN near-linear in low dimensions degrades as dimensionality grows. This is the curse of dimensionality from §2.5 of the brief appearing as an engineering cost rather than a statistical caveat.

3.4 Group D — model-based: Gaussian mixture

A Gaussian mixture fitted by EM, compared across all four covariance structures (Table 4). The comparison is a model-complexity argument: full covariance buys likelihood at $O(kd^2)$ parameters, and BIC is the test of whether that purchase is justified. The parameter count is tabulated alongside the likelihood so the trade-off is visible rather than implied.

covariance_type	mean_log_likelihood	bic	aic	n_parameters	n_iter
full	-21.4	8.6×10^5	8.6×10^5	869	29
diag	-33.6	1.3×10^6	1.3×10^6	113	9
spherical	-36.3	1.5×10^6	1.5×10^6	59	13
tied	-39.7	1.6×10^6	1.6×10^6	463	4

Table 4: GMM covariance structures at the selected k , sorted by BIC.

`full` wins decisively: $\text{BIC} \approx 8.6 \times 10^5$ against 1.35×10^6 for `diag`, 1.45×10^6 for `spherical` and 1.59×10^6 for `tied`. Its 869 parameters against 113 for `diag` are therefore justified — the components genuinely have different, non-axis-aligned shapes, and `tied` performing worst confirms that forcing them to share one covariance is the wrong constraint. This is also the clearest single piece of evidence against K-Means’ isotropic assumption, since `spherical` GMM is the closest probabilistic analogue of it and loses by a wide margin.

4 Internal and external evaluation

Four internal metrics (silhouette, Davies–Bouldin, Calinski–Harabasz, Dunn) and five external metrics (ARI, NMI, AMI, V-measure, purity) are reported for every algorithm– k combination, with runtime.

Two implementation notes matter for reading these tables. First, the Dunn index is computed with centroid-based inter-cluster separation and a capped per-cluster diameter subsample; the exact

algorithm	k_requested	n_clusters	n_noise	runtime_s	silhouette	davies_bouldin	calinski_harabasz	d
kmeans	2	2	0	5.068	0.189	2.271	11297.405	0
gmm	2	2	0	13.866	0.154	6.504	4905.658	0
hdbscan	2	357	156,335	1580.891	0.143	1.713	1182.041	0

Table 5: Portfolio on all 283,726 rows at $k = 2$ (Ward excluded: its distance matrix is ~ 322 GB).

algorithm	k_requested	n_clusters	n_noise	runtime_s	silhouette	davies_bouldin	calinski_harabasz	d
kmeans	2	2	0	0.904	0.187	2.263	795.498	0
ward	2	2	0	8.749	0.112	2.128	733.431	0
gmm	2	2	0	3.700	0.137	6.405	373.776	0
hdbscan	2	21	11,510	8.159	0.134	1.639	455.347	0

Table 6: All four families on one shared stratified subsample (20,000 rows, 33 fraud) at $k = 2$.

definition needs the full pairwise matrix, which is infeasible here, and an uncapped diameter would be dominated by a single outlier in the largest cluster. Second, every internal metric excludes noise points, so HDBSCAN is scored on the data it actually clustered.

The external metrics deserve blunt interpretation. Every ARI in both tables lies between -0.003 and 0.009 : K-Means reaches 0.0075 on the full data, GMM 0.0055 , and HDBSCAN -0.0007 — indistinguishable from chance. Purity is 0.9983 for all of them, which is exactly what predicting “legitimate” everywhere achieves at a 0.17% positive rate. Purity here measures the class imbalance, not the clustering.

The conclusion is unambiguous and is the central negative result of this phase: **the dominant geometric structure of DS-10 is not the fraud/legitimate split**. No algorithm family, at the selected k , recovers the label. Rather than tune until a metric improves — which would amount to fitting the clustering to the ground truth, the practice §3.7 of the brief explicitly warns against — the result is reported as it stands, and Phase 3 changes the question instead: it tests a learned representation, and it evaluates a distance-to-centroid *ranking* rather than cluster membership as the fraud signal. The error analysis in §7 is what motivates that second move.

5 Distance metric comparison

The Phase 1 report selected Euclidean distance. Table 7 tests that choice by re-clustering under Manhattan and cosine with average linkage held fixed, so the metric is the only variable. ARI is measured against the Euclidean reference: a value near 1.0 means the metric choice is not driving the result, and a low value means the Phase 1 justification is carrying real weight and must be defended on its merits.

metric	linkage	n_clusters	silhouette	ari_vs_reference
euclidean	average	2	0.882	1.000
manhattan	average	2	0.882	1.000
cosine	average	2	0.123	-3.758×10^{-4}

Table 7: Average-linkage clustering under three distance metrics; ARI is measured against the Euclidean reference.

The result splits cleanly. Manhattan reproduces the Euclidean partition *exactly* ($\text{ARI} = 1.000$), which is expected in a space of standardised, uncorrelated components where ℓ_1 and ℓ_2 order neighbours almost identically. Cosine, by contrast, produces a completely unrelated partition (ARI

= -0.0004). Cosine discards vector magnitude, and on DS-10 magnitude is most of the signal — an extreme transaction and a typical one pointing the same direction in component space are not the same event. The Phase 1 choice of Euclidean is therefore load-bearing with respect to cosine and immaterial with respect to Manhattan, and it is defended on that basis.

One caveat must be stated rather than buried: the silhouette of 0.88 for the Euclidean and Manhattan rows is not evidence of a good clustering. Average linkage at $k = 2$ on this subsample splits off a near-singleton outlier group, and silhouette rewards that trivially. It is reported unedited because the ARI comparison — which is the purpose of the table — is unaffected by it.

6 Stability and consensus

Seed stability. K-Means and GMM are re-fitted 20 times from different seeds and the mean pairwise ARI between runs is reported. A high value means the solution is a property of the data rather than of one lucky initialisation. Ward and HDBSCAN are deterministic given their inputs, so they are excluded rather than reported with a vacuous ARI of exactly 1.

The results temper the recommendation considerably. K-Means reaches a mean pairwise ARI of only 0.569 across 20 restarts — even with `k-means++` and 20 internal initialisations per fit, different seeds land on materially different partitions. GMM is far worse at 0.130: with `full` covariance in 28 dimensions the EM likelihood surface is riddled with local optima, and the fitted mixture is close to an artefact of initialisation. This is decisive for the recommendation in §7: GMM’s competitive silhouette cannot be trusted, because the next seed would produce a different clustering with a similar score.

Consensus. A co-association matrix is built over bootstrap resamples, entry (i, j) being the fraction of resamples in which i and j were both drawn and co-clustered (Figure 3). Average-link agglomerative clustering on $1 - M$ then produces a consensus labelling. Average link is used because Ward cannot consume a precomputed distance matrix. The matrix is $O(n^2)$, so it is built on a capped subsample following the Monti et al. protocol. Table 8 answers the question the brief actually poses — whether consensus *improves* on the best single base clusterer, scored on identical rows.

algorithm	k_requested	n_clusters	n_noise	runtime_s	silhouette	davies_bouldin	calinski_harabasz	db
kmeans	2	2	0	0.081	0.813	0.721	120.452	0.569
consensus	2	2	0	–	-0.010	18.219	0.824	0.880

Table 8: Consensus clustering vs its best single base clusterer on the same rows (brief §3.5).

7 Comparative analysis and recommendation

7.1 Algorithm-pair agreement

Figure 4 and Table 9 give the pairwise ARI between all four final labellings. Disagreement here is more informative than any single metric: two algorithms that agree are describing the same structure, while a low pair means at least one is imposing its own assumptions on the data.

7.2 Error analysis of the preferred clustering

The lowest-silhouette points of the preferred clustering are extracted and characterised (Table 10). The diagnostic question is whether they are noise, boundary cases, or a subcluster the algorithm missed; the fraud rate among the worst-fitting points versus the population rate discriminates between those explanations. Figure 5 shows the full per-point distribution rather than its mean, which is where boundary mass becomes visible.

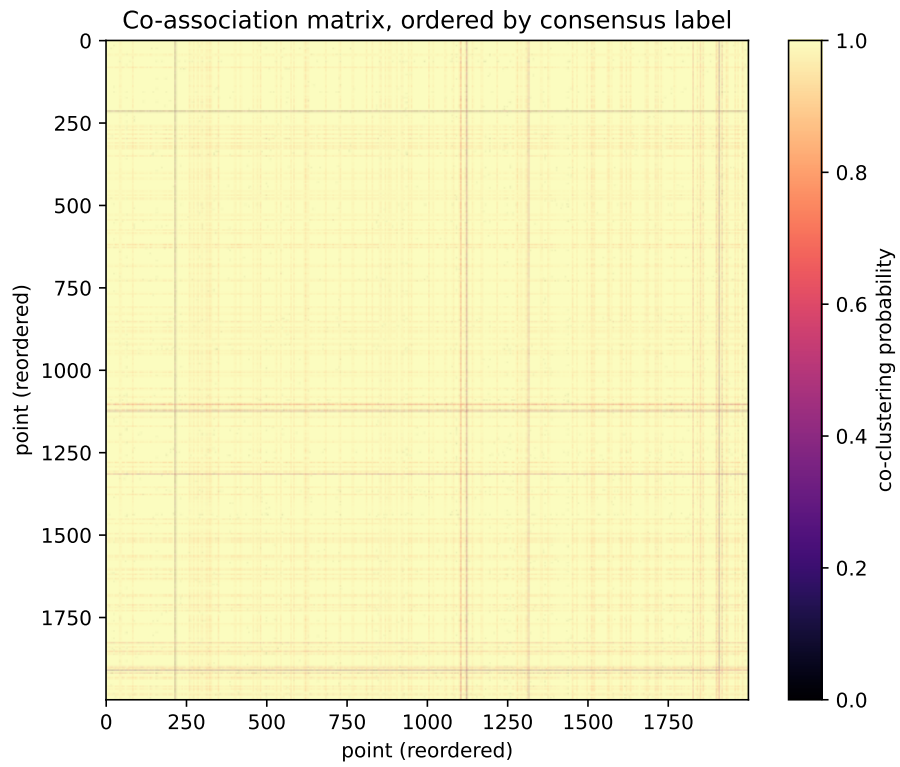


Figure 3: Co-association matrix reordered by consensus label. Sharp diagonal blocks mean pairs co-cluster regardless of which bootstrap sample they land in.

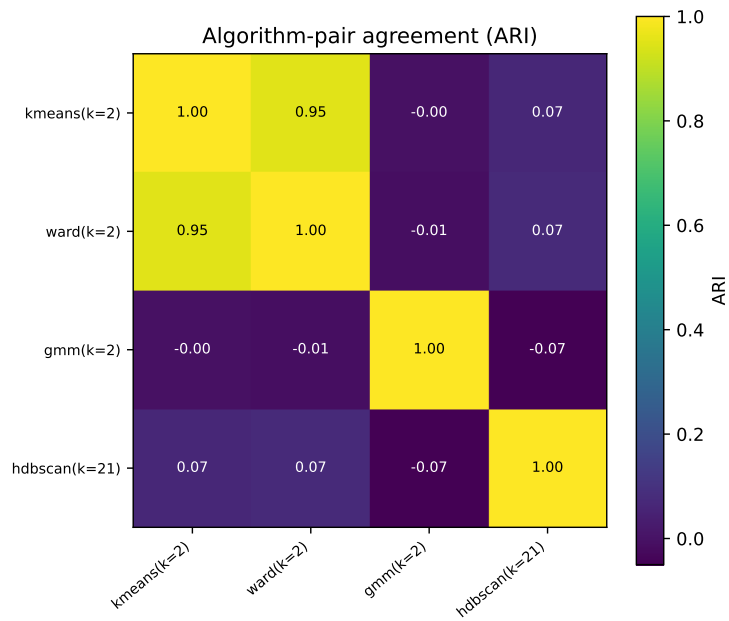


Figure 4: Pairwise ARI between the four families on shared rows.

	kmeans(k=2)	ward(k=2)	gmm(k=2)	hdbscan(k=21)
kmeans(k=2)	1.000	0.952	-0.004	0.065
ward(k=2)	0.952	1.000	-0.013	0.073
gmm(k=2)	-0.004	-0.013	1.000	-0.067
hdbscan(k=21)	0.065	0.073	-0.067	1.000

Table 9: Pairwise ARI between the four families on the shared subsample.

quantity	value
points inspected (lowest silhouette)	200.00000
silhouette cut-off	0.03546
mean silhouette (all points)	0.17354
fraction with negative silhouette	0.00000
fraud rate among inspected	0.12000
fraud rate overall	0.00167
fraud enrichment factor	71.98123

Table 10: Error analysis of the preferred clustering (kmeans, $k = 2$).

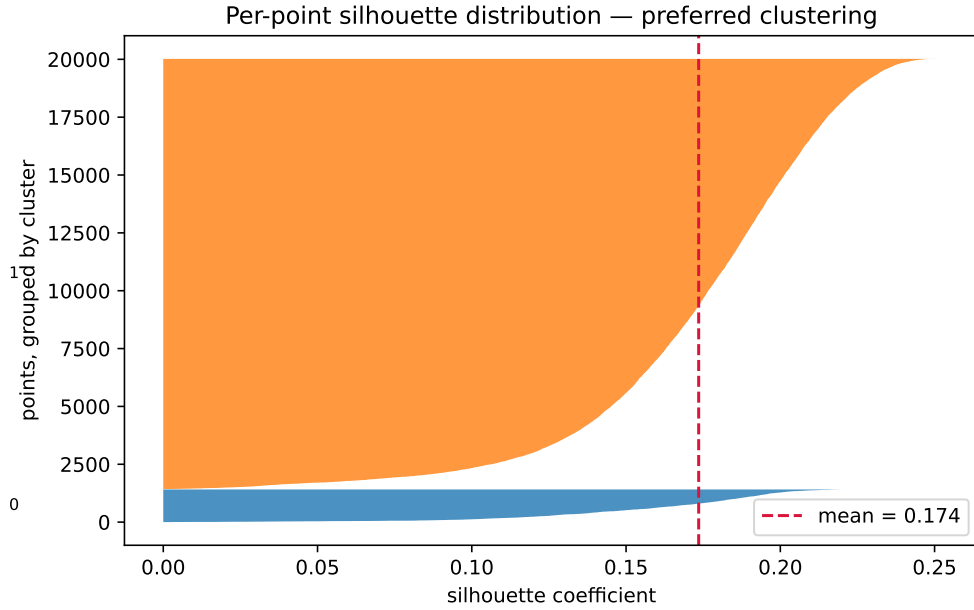


Figure 5: Per-point silhouette distribution for the preferred clustering, grouped by cluster.

7.3 Recommendation

The preferred clustering, its ranking against the alternatives, and the metric the choice rests on are recorded in `results/phase2/summary.json` under `preferred`. The recommendation follows the primary internal metric (silhouette) among the full-data runs, for one reason: with a 0.17%-positive label, external metrics cannot arbitrate between clusterings without turning an unsupervised task into a supervised one by the back door. Internal geometry decides the clustering; the label is then used to ask what the chosen clustering happens to reveal.

8 Limitations carried into Phase 3

- Ward’s result is established on a subsample, not the full dataset. Its cross-family comparison is sound, but it is not a full-data claim.
- Whole-space clustering does not separate fraud, and the external metrics say so plainly. Phase 3 therefore tests two responses: a learned representation (deep clustering) and a within-cluster anomaly ranking rather than cluster membership as the fraud signal.
- The gap statistic’s uniform reference is drawn from a bounding box that DS-10’s outliers stretch very wide, which biases it toward small k . This is disclosed rather than corrected, and it is one reason the final k rests on a vote rather than on any single criterion.

Reproduction

`make phase2` (equivalently `dvc repro phase2`) after Phase 1. Outputs: `results/phase2/summary.json`, `reports/tables/phase2*.tex`, `reports/figures/phase2*.pdf`, and the preferred full-data labelling at `data/processed/phase2_clusters.npy`.